

1st INTERNAL/ OPERATING SYSTEM/ PCC-CS504 / 5TH SEM-CSE/16.9.2025/ FM: 15

Answer any 3

1. Describe storage hierarchy with a suitable diagram. What is scheduler? 3+2=5
2. Describe the life cycle of a process with a process state diagram. What are CPU Burst and I/O Burst? 3+2=5
3. What is PCB? What are the information's stored in a PCB? What is context switching? 1+3+1=5
4. What is FCFS scheduling? Suppose that the processes arrive in the order: P_1, P_2, P_3 with CPU Burst time of 18, 6, 3. With a Gantt chart find waiting time of every process and also find average waiting time. 1+2+2=5

2nd INTERNAL/OPERATING SYSTEM/ PCC-CS504 / 5TH SEM-CSE/20.12.2025/ FM: 15

Answer any 3

1. What is Thread? Write the difference between Process and Thread? 3+2=5
2. Draw the diagram of paging hardware and discuss how it works? 3+2=5
3. Discuss the difference between Paging and Segmentation in details. 5
4. Discuss different address binding scheme in details? Out of this which is mostly used and why? 5

[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ DEPT -CSE/ PAPERCODE-PCC-CS-502

Subject: Discrete Mathematics Time 3/4 hour. Answer All $3 \times 5 = 15$

1. Obtain the conjunctive normal form of $\{q \vee (p \wedge r)\} \wedge \sim \{(p \vee r) \wedge q\}$. OR

$\checkmark (p \vee q) \wedge (\sim p \wedge (\sim q \vee \sim r)) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology.

2. $\checkmark$ Solve $y_{n+2} + 3y_{n+1} + 2y_n = 3^n$. OR

Consider the statements p: "I will study discrete mathematics", q: "I will go to movie", r: "I am in a good mood". Write the following statements in terms of p, q, r and logical connectives. Construct the truth table. "I will not go to a movie and I will study discrete mathematics if and only if I am in a good mood."

3. $\checkmark$ Develop the recurrence relation for the Fibonacci sequence of numbers $\{1, 1, 2, 3, 5, 8, 13, \dots\}$ And hence solve it. OR $\times$ Obtain DNF of $p \rightarrow (p \rightarrow q) \wedge \{\sim(\sim q \vee \sim p)\}$.

2nd Class Test Dept. of CSE, J G E C December 2025

Discrete Mathematics PC C-CS 501 Answer any Three. FM. $3 \times 5 = 15$

- $\checkmark$ In a group of 100 students, 72 students can speak English and 43 students can speak Hindi. Based on these data, answer the following questions:

- (a) Find the number of students who can speak English only.
(b) Find the number of students who can speak Hindi only.
(c) Find the number of students who can speak both English and Hindi.

$\checkmark$ State Euclid's Algorithm. Using this Algorithm find the G C D (1479, 272).

$\checkmark$ Find the total number of distinct divisors of 444720375.

$\checkmark$ Find the missing digit in the number 8630480*874 if it divisible by i) 11, ii) 13.

5. Let $A, B, C \in R$ Consider the function $f: A \rightarrow B$ and $g: B \rightarrow C$ defined by $f(a) = 2a+1$, $g(b) = b/3$ verify $(g \circ f)^{-1} = \{f \circ g\}^{-1}$.

$$\begin{array}{r} 13 \overline{) 2986} \quad 12 \\ \underline{26} \\ 36 \\ \underline{26} \\ 10 \\ \underline{13} \\ 27 \\ \underline{26} \\ 1 \end{array}$$

Full marks: 15

(You can answer all questions but maximum score will be 15)

- 7) Give example of the following i) meta data, ii) derived attribute, iii) composite attribute and iv) aggregate function. 2
- 8) Consider two relations: Instructor(ID, Name, Dept, Salary) and Teaches(ID, CourseID, SectionID, Sem, Year). Assume ID is the foreign key attribute of Teaches referencing Instructor relation. What happens if you want to insert a record into Teaches and delete a record from the Instructor relations, respectively? 2

Students

RollNo	Sem	Paper	SGPA
15R01	I	SE	9
17R05	III	AI	9
16R05	II	DBMS	7
15R01	I	AI	8
17R05	III	SE	9
16R01	I	AI	8
15R05	II	OS	6
17R05	III	OS	7
15R01	I	OS	7
16R01	I	SE	9

Qualifier

Paper	SGPA
SE	9
AI	8
OS	7

- 3) Consider the relations, Students and Qualifier, and show the output instance for T1, T2, and T3. $1+1+2=4$

$$T1 = \Pi_{\text{RollNo, Sem}}(\text{Students}) \times \text{Qualifier}$$

$$T2 = \Pi_{\text{RollNo, Sem}}(T1 - \text{Students})$$

$$T3 = \Pi_{\text{RollNo, Sem}}(\text{Students}) - \Pi_{\text{RollNo, Sem}}((\Pi_{\text{RollNo, Sem}}(\text{Students}) \times \text{Qualifier}) - \text{Students})$$

- 4) There are two relations, Instructor(ID, Name, Dept, Salary) and Teaches(ID, CourseID, SectionID, Sem, Year) having n and m numbers of rows, respectively. Each tuple or row of Instructor and Teaches needs si and st bytes to store themselves, respectively. Say, there are k numbers of instructors from the physics department and p number of instructors with salaries greater than 90000. Say, each comparison takes t seconds. Find the size of all the results and comparison time for execution of the queries. $1.5 \times 4 + 2 = 8$

a) Instructor x Teaches b) Instructor $\bowtie_{\text{Instructor.ID = Teaches.ID}}$ Teaches c) $\sigma_{\text{Dept = "Physics" \wedge Salary > 90000}}(\text{Instructor} \bowtie_{\text{Instructor.ID = Teaches.ID}} \text{Teaches})$

d) $\sigma_{\text{Dept = "Physics" \wedge Salary > 90000}}(\text{Instructor} \bowtie_{\text{Instructor.ID = Teaches.ID}} \text{Teaches})$

e) Left outer join and right outer join of Instructor and Teaches (find only the size of the resultant instances while I)

Teaches.ID is the foreign key referencing Instructor, ii) Teaches.ID is not a foreign key attribute)

- 5) Consider the relations: R(A,B,C,D), S(E,F,G), T(G,D,H) and draw the equivalent query tree of the following query and optimize it. $2+2=4$

$$\Pi_{A,B,H}(\sigma_{G=T.G \wedge D=T.D}(R \times S \times T))$$

DBMS (Class Test II/ PCC-CS503)

Department of Computer Science and Engineering
2025

Time: 45 min

Based

Full marks: 15

Answer any three

- 1) Describe ACID property of a transaction with examples. Define serializable schedule. $3+1=4$
- 2) Consider a relation R(Model, Year, Price, Manufac_plant, Color) with the following set of F of functional dependencies: $F = \{\text{Model} \rightarrow \text{Manufac_plant}, \{\text{Model}, \text{Year}\} \rightarrow \text{Price}, \text{Manufac_plant} \rightarrow \text{Color}\}$
- d) Find the candidate key.
- e) Determine whether the relation is in 3NF or BCNF with proper justification.
- f) Say, R is decomposed in R1(Model, Year, Price) and R2(Model, Manufac_plant, Color). Is this decomposition lossless? $3+3+2=8$
- 3) Describe the state transition diagram of transaction execution. 3

$$\begin{array}{r} 3 \overline{) 2886} \quad 12 \\ \underline{26} \\ 26 \\ \underline{26} \\ 0 \end{array}$$

1st Internal Assessment Examination (2025)
Dept: CSE Semester: 5th
Sub: Software Engineering (PCC-CS501)

Answer all questions

1. Explain prototyping model? Under what circumstances is it beneficial to construct a prototype? 3+1
2. Define project size in terms of software metrics. 2
3. Write down the role of SRS documents in the life cycle model of software development. 2
4. What do you mean by software risk? Explain different types of risks with examples? 1+3
5. A particular real-time system is estimated to contain 11×10^7 LOC. Using Intermediate COCOMO, find out the total development time and effort. Given that it will require a very high PCAP rating(0.70), very high MODP rating(0.82), extra high TIME constraints(1.66), very high TURN rating(1.15), considering all other ratings to be nominal. 3

2nd Internal Assessment Examination (2025)

Dept: CSE Semester: 5th Sub: Software Engineering (PCC CS501)

1. Write down the criteria for good software design. Explain different types of cohesion with examples. 2+32.
 2. What do you mean by balancing a DFD? Give suitable example. Explain different levels of testing with their goals. 2+3
 3. Write down the differences between Code Walkthrough and Code Inspection. Draw Control Flow graph for the following code and calculate cyclomatic complexity. 2+3
- ```
1 IF X = 300
2 THEN IF Y > Z
3 THEN X = Y
4 ELSE X = Z
5 END IF
6 END IF
7 PRINT X
```

Operation Research PEC-CS501D CSE, 3rd Year, 1<sup>st</sup> Semester, 2025 Class test 1, Full Marks-15, Time- 45 minutes

(1). Solve the given transportation table using N/W Corner, Least Cost Method and VAM. Calculate the transportation cost in each 3X3=9

|             | 1  | 2  | 3  | 4  | Availability |
|-------------|----|----|----|----|--------------|
| A           | 7  | 3  | 5  | 5  | 34           |
| B           | 5  | 5  | 7  | 6  | 15           |
| C           | 8  | 6  | 6  | 5  | 12           |
| D           | 6  | 1  | 6  | 4  | 19           |
| Requirement | 21 | 25 | 17 | 17 |              |

or Use simplex method to solve the following LPP

Maximize  $Z = x_1 + x_2 + 3x_3$

Subject to constraints

$$3x_1 + 2x_2 + x_3 \leq 3$$

$$2x_1 + x_2 + 2x_3 \leq 2$$

$$x_1, x_2, x_3 \geq 0$$

2. Use the graphical method to solve the following linear programming problem 6

Maximize  $Z = 15x_1 + 10x_2$

Subject to constraints,

$$4x_1 + 6x_2 \leq 360$$

$$3x_1 \leq 180$$

$$5x_2 \leq 200$$

$$x_1, x_2 \geq 0$$

## Assessment E Semester End

Operation Research PEC-CS501D CSE, 3rd Year, 1<sup>st</sup> Semester, Class test 2, Full Marks-15, Time- 45 minutes, Date: 19.12.25

1. Draw the network diagram and determine the critical path for the given data. 3+6

| Job      | 1-2 | 1-3 | 2-4 | 3-4 | 3-5 | 4-5 | 4-6 | 5-6 |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|
| Duration | 6   | 5   | 10  | 3   | 4   | 6   | 2   | 9   |

OR 2. Using the following cost matrix, determine (a) optimal job assignment (b) total cost of assignments 8+1

| Mechanic |   | JOB |    |    |    |    |
|----------|---|-----|----|----|----|----|
|          |   | 1   | 2  | 3  | 4  | 5  |
|          | A | 30  | 25 | 33 | 35 | 36 |
|          | B | 23  | 29 | 38 | 23 | 26 |
|          | C | 30  | 27 | 22 | 22 | 22 |
|          | D | 25  | 31 | 29 | 27 | 32 |
|          | E | 27  | 29 | 30 | 24 | 32 |

3. Solve the game whose pay-off matrix is given by. Is the game fair? 5+1

Player B

|    | B1 | B2 | B3 |
|----|----|----|----|
| A1 | 6  | 8  | -1 |
| A2 | 2  | 3  | 5  |
| A3 | 14 | 18 | 12 |

Player A



# 1<sup>st</sup> Internal Assessment Examination (2025)

Dept: CSE Semester: 5<sup>th</sup>

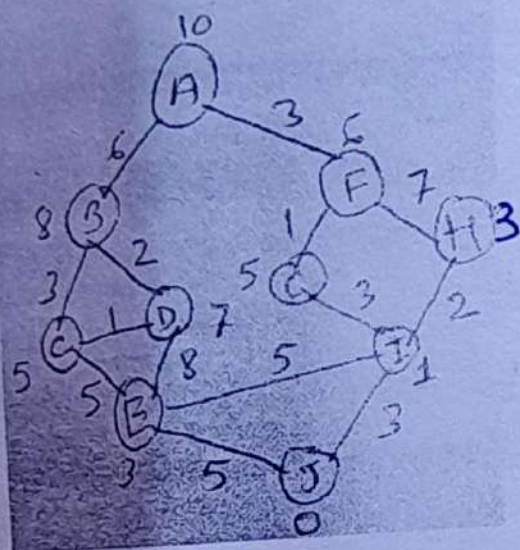
Sub: Artificial Intelligence (PEC-CS502C)

Answer all questions

1. Explain the term 'Environment' in the context of artificial intelligence. Explain three properties of the AI environment with examples. 1+3

2. Write down the properties of a search algorithm. Write down the differences between informed search and uninformed search. Using A\* search algorithm, find the most cost-effective path to reach from start state A to final state J for the graph given below. 2+3+3

3. Explain the functions of different components of the learning agent with a proper diagram. 3



## 2<sup>nd</sup> Internal Assessment Examination (2025)

Dept: CSE Semester: 5<sup>th</sup> Sub: Artificial Intelligence (PEC-CS502C)

1. Define reasoning? Differentiate between deductive reasoning and inductive reasoning. Solve the following Crypt Arithmetic problem: EAT + THAT = APPLE 1+2+3
2. Convert the following proposition into FOL : (a) All cats are mammals (b) All mammals are animals (c) Kitty is a cat. Using forward chaining proof that Kitty is an animal 3+2
3. Consider the game tree given below and apply Alpha-Beta pruning algorithm on it. What is the time complexity of Alpha-Beta algorithm. 3+1

